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Bacteriology and Clinical Significance of Hemolytic *Haemophilus* in the Throat

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Hemolytic *Haemophilus* are rarely isolated in the clinical laboratory, as they do not grow on sheep or human blood-agar alone. On rabbit blood-agar they grow well and are hemolytic, but they grow less well and are not hemolytic on sheep blood-agar with added X and V factors. A survey was made to determine their incidence in pharyngitis. From 28 of 100 sore throats and from 57 of 100 normal throats, only normal bacterial flora were isolated. β-Streptococci were present in significant numbers in 9 sore and 11 normal throats; staphylococci in 8 sore and 4 normal throats; pneumococci in 20 sore and 11 normal throats; *H. influenzae* or *H. parainfluenzae* in 13 sore and no normal throats; hemolytic *Haemophilus* in 30 sore and 18 normal throats; enteric bacilli in 1 of each; *Candida* and *Neisseria* in 2 sore throats each. All of the 33 hemolytic *Haemophilus* isolates identified to species were *H. parahaemolyticus*. All were sensitive in vitro to chloramphenicol, erythromycin, and tetracycline; 30 were sensitive to ampicillin and 30 to penicillin, 26 to novobiocin, and 12 to methicillin. *H. influenzae*, *H. parainfluenzae*, and *H. haemolyticus* are indistinguishable by Gram stain morphology, but *H. parahaemolyticus* is larger than the other three. Hemolytic and nonhemolytic species are indistinguishable by colonial morphology or by nutritional requirements; only hemolysis gives positive differentiation. Nevertheless, only rarely would this be of clinical importance. *H. parahaemolyticus* apparently may cause pharyngitis, but it is almost always susceptible to penicillin and rarely if ever causes sequelae.

In 1919, Pritchett and Stillman (12) reported "Bacillus X," apparently a hemolytic *Haemophilus*, from the upper respiratory tract. Jawetz et al. (5) reported that *H. haemolyticus* "occurs both in the normal nasopharynx and associated with rare upper respiratory-tract infections of moderate severity in children." Henriksen (4) reported the isolation of what was apparently *H. parahaemolyticus* from the throat of a patient with recurrent sore throats and from 8 of 44 medical students presumably with normal throats. Kimler (6) mentioned *H. haemolyticus* being found in stools, but a search of the literature has revealed no other report of the isolation of hemolytic *Haemophilus* from pneumonia or from wounds, abscesses, meningitis, ear infections, peritonitis, or any other extrarespiratory infections.

DeLeeuw-Frans et al. (1) reported on 18 strains of hemolytic *Haemophilus* isolated from respiratory tracts in Belgium. Only one was *H. haemolyticus*, recovered from a normal throat; the other 17 were *H. parahaemolyticus*, all from cases of pharyngitis. Because of the scarcity of recent information on the subject in the United States, a survey was made to determine whether the hemolytic *Haemophilus* should be considered pathogenic in the throat.

MATERIALS AND METHODS

In November and December 1966, swabs were taken of 100 clinically normal and 100 sore throats, mostly from hospital employees and adult patients. The "normal throat" group and 8 of the 100 "sore throat" group were composed of hospital laboratory employees, none of whom admitted to recent antimicrobial therapy, even lozenges. These cultures were taken by the author. Of the remaining 92 people with sore throats, 7 were inpatients cultured by their physicians and 85 were outpatients who admitted to no recent antimicrobial therapy; the latter were cultured by several people trained by the author.

The inocula were swabbed onto 5% rabbit blood tryptic soy agar (TSA; BBL) and streaked for isolation. An optochin disc (Taxo P; BBL) was put on the swabbed area of each culture plate, and the cultures were incubated overnight. β-Hemolytic colonies were Gram-stained, and in 33 cases (17 from normal and 16 from sore throats) hemolytic *Haemophilus* were streaked out for further testing. This consisted of speciation by use of X and V factor-impregnated paper strips (Flu-Strips; Case Laboratories, Chicago,

Ill.) on TSA, and in vitro disc sensitivity testing on 5% rabbit blood agar with the antimicrobials used routinely in our laboratory for gram-positive isolates. The gram-positive set was used because it is our experience that acute pharyngitis is usually treated empirically with an antimicrobial useful against β-hemolytic streptococci.

Cultural characteristics of *H. parahaemolyticus* were investigated by use of the X and V factor strips, a hemolytic, pigmented, coagulase-positive *Staphylococcus*, and rabbit and sheep blood-agars and TSA.

RESULTS

From 28 of 100 sore throats and from 57 of 100 clinically normal (i.e., not sore) throats, only normal bacterial throat flora were isolated. Bacterial flora found in normal and sore throats are summarized in Table 1. *Mycoplasma* cultures were not made. The term "significant numbers" is used when 10% or more of the flora cultured was the organism in question. Margileth and Mella (8) considered 5 to 10 colonies of β-hemolytic streptococci significant and more than 25 to 30 colonies a severe infection, but that does not take into account the total number of colonies.

There appeared to be as many carriers of β-hemolytic streptococci as there were patients, but isolates of staphylococci, pneumococci, and *Haemophilus* were definitely more numerous from sore throats. This was most noticeable with the nonhemolytic *Haemophilus* species (*H. influenzae* or *H. parainfluenzae*; these were not identified to species), all of which came from sore throats. Pneumococci, staphylococci, and hemolytic *Haemophilus* were found in significant numbers almost twice as often in sore throats as in normal throats. Table 2 lists possible pathogens by percentage of the total throat flora. In 3 normal and 11 sore throats, there was more than one possible pathogen in significant numbers.

TABLE 1. *Bacterial flora in normal and sore throats*

Possible pathogen	No. of cultures (= per cent)	
	Normal throats	Sore throats
<10% of any one	57	28
≥10% β-Hemolytic *Streptococcus*	11	9
Staphylococcus[a]	4	8
Diplococcus pneumoniae	11	20
Nonhemolytic *Haemophilus*[b]	0	13
Hemolytic *Haemophilus*	18	30
Enteric gram-negative bacilli	1	1
Candida	0	2
Neisseria	0	2

[a] Hemolytic or coagulase-positive, or both.
[b] *H. influenzae* or *H. parainfluenzae*.

TABLE 2. *Amounts of possible pathogens*

Organism	Percentage of flora	No. of cultures		
		Normal throats	Sore throats	Total
β-Hemolytic streptococci	10–30	9	7	16
	40–60	2	2	4
	70–100	0	0	0
	Totals	11	9	20
Staphylococcus	10–30	3	5	8
	40–60	0	2	2
	70–100	1	1	2
	Totals	4	8	12
Pneumococcus	10–30	8	13	21
	40–60	2	7	9
	70–100	1	0	1
	Totals	11	20	31
Nonhemolytic *Haemophilus*	10–30	0	12	12
	40–60	0	1	1
	70–100	0	0	0
	Totals	0	13	13
Hemolytic *Haemophilus*	10–30	12	24	36
	40–60	5	3	8
	70–100	1	3	4
	Totals	18	30	48
Enteric bacilli	10	1	1	2
Candida	10	0	1	1
	100	0	1	1
	Totals	0	2	2
Neisseria	40–60	0	1	1
	70–100	0	1	1
	Totals	0	2	2
Totals		45	85	130

All 33 isolates of hemolytic *Haemophilus* which was identified to species were *H. parahaemolyticus*, requiring only the V factor. This was confirmed by satellitism around *Staphylococcus* on TSA.

Results of the sensitivity tests are shown in Table 3. There was no obvious relationship among sensitivities to the three penicillins. Two isolates were resistant to all three, and 12 were sensitive to all three. These results confirm those of De-Leeuw-Frans et al. (1), who reported that *H. parahaemolyticus* is generally more sensitive to penicillin and much less sensitive to novobiocin than is *H. influenzae*, with methicillin sensitivity very variable.

TABLE 3. *In vitro sensitivity of 33 Haemophilus parahaemolyticus isolates*

Antimicrobial agent	No.	Percent
Ampicillin (10 μg)	30	90.9
Chloramphenicol (5 μg)	33	100
Erythromycin (2 μg)	33	100
Methicillin (5 μg)	12	36.4
Novobiocin (5 μg)	26	78.8
Penicillin (2 units)	30	90.9
Tetracycline (5 μg)	33	100

The cultures confirmed that *H. parahaemolyticus* does not grow on sheep blood-agar alone; and there was only scanty growth around the *Staphylococcus*. It was also confirmed that *H. parahaemolyticus* is not hemolytic on sheep blood-agar (9).

DISCUSSION

The hemolytic *Haemophilus* species do not grow on human or sheep blood-agar, and rarely are considered virulent enough to warrant culture on rabbit (or guinea pig or rat; 7) blood-agar. In our laboratory, where rabbit blood-agar is used routinely, we see many hemolytic *Haemophilus* organisms in throat cultures.

DeLeeuw-Frans et al. (1) suggested that *H. parahaemolyticus* causes pharyngitis, as they did not find it in clinically normal throats. This survey, however, indicates that it may be present in significant numbers without causing pharyngitis.

Among those people with sore throats, three times as many had hemolytic *Haemophilus* as had β-streptococci in appreciable numbers, and there were twice as many with hemolytic as with nonhemolytic *Haemophilus*. This is not to imply that either species of *Haemophilus* is more important than β-hemolytic *Streptococcus*; quite the contrary is true. Similarly, of the two kinds of *Haemophilus*, the nonhemolytic species undoubtedly are more important than the hemolytic from the standpoint of possible sequelae. No sequelae have been reported in the literature from hemolytic *Haemophilus* infections, except perhaps its rare occurrence in subacute endocarditis (11), and none has been seen in this hospital in the 20 years since rabbit blood-agar has been used.

As with *H. influenzae* and *H. parainfluenzae*, differentiation between the hemolytic species may be made by X (iron porphyrin) and V (nicotinamide adenine dinucleotide-nicotinamide adenine dinucleotide phosphate) factor requirements. *H. haemolyticus* requires both; *H. parahaemolyticus* requires only the V factor. In her description of *H. parahaemolyticus* as a new species, Pittman (11) stated that it frequently is associated with acute pharyngitis and occasionally causes subacute bacterial endocarditis. Until her differentiation of the two species, both were called *H. haemolyticus*; this may account for statements in textbooks that *H. haemolyticus* is found in upper respiratory tract infections and sometimes in subacute bacterial endocarditis. Where the hemolytic species have been identified, *H. parahaemolyticus* has been more common; Fildes (2) reported 13 of 14 isolates were *H. parahaemolyticus*; Valentine and Rivers (13) reported 10 of 15; DeLeeuw-Frans et al. (1) reported 17 of 18; and this survey showed 33 of 33.

Haemophilus species usually are identified in clinical laboratories only by their X and V factor requirements, by use of a colorless agar or chocolate-agar; on neither of these media is hemolysis visible. Without demonstration of hemolysis, it is possible that the hemolytic *Haemophilus* species could be wrongly identified as *H. influenzae* or *H. parainfluenzae*. In Gullekson's report (3) of *H. parainfluenzae* in neonatal meningitis, for example, the gram-negative bacillus isolated is described as (i) growing on chocolate-agar but not sheep blood-agar, (ii) not reacting with antisera to the six serotypes of

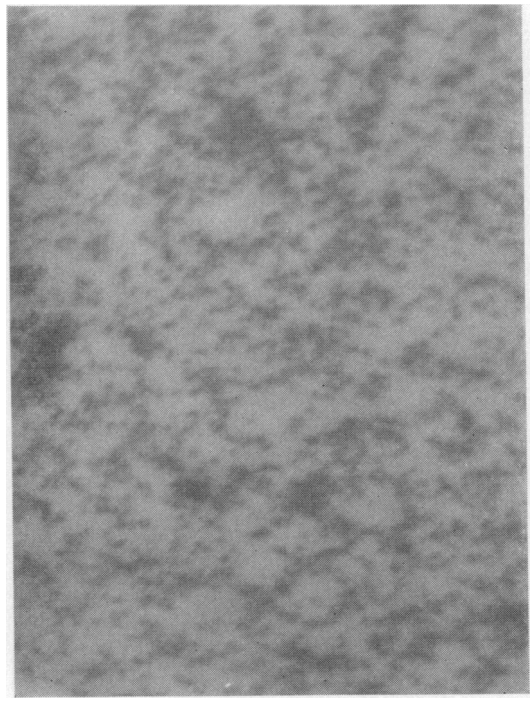

FIG. 1. *Haemophilus influenzae. Gram stain.* × *1,500.*

H. influenzae, and (iii) not requiring the X factor, determined by its growth around *S. aureus* on nutrient agar. None of these criteria excludes *H. parahaemolyticus*, although Gram stain may have. (Our laboratory has never isolated hemolytic *Haemophilus* from spinal fluid.)

The descriptions in *Bergey's Manual* of the colonies of hemolytic *Haemophilus* differ only in that those of *H. haemolyticus* are "transparent" and those of *H. parahaemolyticus* are "slightly opaque." In the absence of hemolysis, colonies of hemolytic and nonhemolytic *Haemophilus* cannot be distinguished. By Gram stain, however, *H. haemolyticus* resembles *H. influenzae* and *H. parainfluenzae* in its small size (Fig. 1), whereas *H. parahaemolyticus* is an irregularly staining and extremely pleomorphic bacillus much larger than *H. parainfluenzae* and often larger than enteric gram-negative bacilli (Fig. 2).

Pritchett and Stillman (12) could not demonstrate pathogenicity of their hemolytic *Haemophilus* for mice, rats, or rabbits, although pig pleuropneumonia due to *H. parahaemolyticus* has been reported (11). Hemolytic *Haemophilus* has not been isolated in our laboratory from wounds, blood, appendicitis, spinal fluid, stool, urine, abscesses, the eye, or even pneumonia, although our routine use of rabbit blood permits its isolation and recognition.

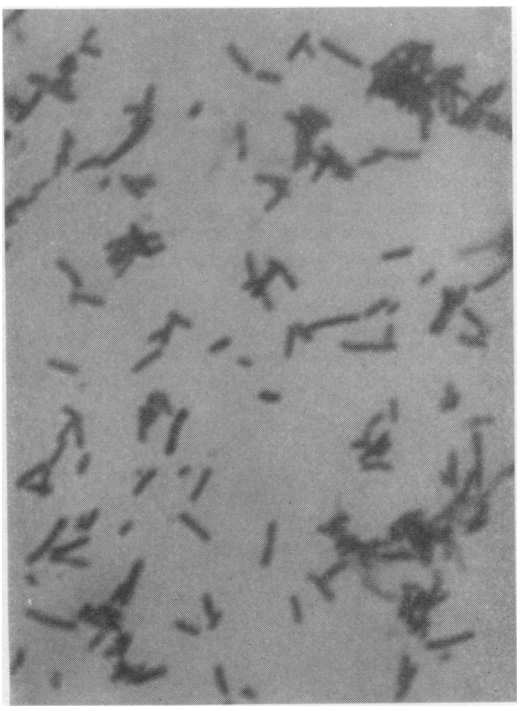

FIG. 2. *Haemophilus parahaemolyticus. Gram-stain.* × *1,500.*

Confusion of hemolytic and nonhemolytic *Haemophilus* species appears, therefore, to be of little clinical importance, except for the possibility of identification of the hemolytic species from upper respiratory-tract cultures as nonhemolytic (more virulent) species. From other clinical material, any *Haemophilus* isolated is almost certain to be nonhemolytic. In addition, *H. parahaemolyticus* would probably yield to an antibiotic therapy regimen devised for *H. influenzae* or *H. parainfluenzae*.

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